Table 4. ADC features

ADC modes/features	ADC1
Resolution	12 bits
Maximum sampling-speed	2.25 Msps
Hardware-offset calibration	X
Single-ended inputs	X
Injected channel conversion	X
Oversampling	up to x1024
Data register	32 bits
DMA support	X
Offset compensation	X
Gain compensation	X
Number of analog watchdogs	3

3.15.1 Analog temperature sensor

The STM32C542xx embed an analog temperature sensor that generates a voltage V_{SENSE} that varies linearly with temperature. The temperature sensor is internally connected to the ADC input channel that is used to convert the sensor output voltage into a digital value.

The sensor provides good linearity but it must be calibrated to obtain a good accuracy of the temperature measurement. As the offset of the temperature sensor varies from chip to chip due to process variation, the uncalibrated internal temperature sensor is suitable for applications that detect temperature changes only.

To improve the accuracy of the temperature sensor measurement, each device is individually factory-calibrated by STMicroelectronics. The temperature sensor factory calibration data are stored by STMicroelectronics in the system memory area, accessible in read-only mode.

3.15.2 Internal voltage reference (V_{REFINT})

The V_{REFINT} provides a stable (bandgap) voltage output for the ADC and the comparator. It is internally connected to ADC input channel.

The precise voltage of V_{REFINT} is individually measured for each part by STMicroelectronics during production test and stored in the system memory area. It is accessible in read-only mode.

3.16 Digital to analog converter (DAC)

The DAC module is a 12-bit, voltage output digital-to-analog converter. The DAC can be configured in 8- or 12-bit mode and can be used in conjunction with the DMA controller. In 12-bit mode, the data may be left-aligned or right-aligned.

An input reference pin, VREF+ (shared with others analog peripherals) is available for better resolution.

The DAC_OUTx pin can be used as general-purpose input/output (GPIO) when the DAC output is disconnected from the output pad and connected to the on chip peripheral. The DAC output buffer can be optionally enabled to allow a high drive output current. An individual calibration can be applied DAC output channel. The DAC output channels support a low-power mode, the sample and hold mode.